QPCODE: 25900871

MAHATMA GANDHI UNIVERSITY, KOTTAYAM, MGU UGP HONOURS
REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION NOVEMBER 2025
FIRST SEMESTER
DSC-MG1DSCICH100- FUNDAMENTALS OF INDUSTRIAL CHEMISTRY
(2024 ADMISSION ONWARDS)

DURATION 1.5 H **MAXIMUM MARKS :50**

PART A

Answer All Questions, Each Question Carries 1 Mark.

1. Which of the following fuels is most likely to cause knocking in a gasoline engine
……………..b. fuel with low octane number
2. What is the main purpose of isomerization process in petroleum refining
… … … … … … **a. To convert normal alkanes to branched alkanes**
3. Which fraction from crude oil is the lightest-… … … … … c-Gasoline
4. Which of the following is the primary element that distinguishes steel from cast iron-
………….a-carbon
5. Rusting of iron is an example of **---------- b- Oxidation-reduction reaction**
6. Which process is used to remove gangue from an ore--------**b. Froth floatation**
7. What is the primary component of clay mineral--------**b- Hydrous Aluminium Silicates**
8. What is the common characteristics of self assembled monolayers (SAMS)
……………… b. They are used to modify surface properties
9. Which term refers to the process where nanoparticles spontaneously organize into a structured form … … … … … … … … … … … **b. self assembly**
10. Which of the following is a characteristics of nanorods--------**a. Length greater than width.**

PART B

Short Answer Questions, Answer Any 4, Each Question Carries 3 Marks

11. What are unit operations in a chemical process

Unit operation is a basic step in a process that involves physical change or chemical transformation during the process like polymerization, separation, evaporation, crystallization, filtration, distillation, isomerization, and other reactions. Eg distillation of acetone and water.

(Definition 2 mark, any example 1 mark)

12. Explain why safety measures are important in a chemical manufacturing plant

Protection of Workers: They ensure that workers are trained and equipped with the necessary skills to handle chemicals safely, reducing the risk of accidents and injuries.

Environmental Protection: Safety measures help prevent the release of hazardous chemicals into the environment, protecting the health of the community and the ecosystem. **Legal Compliance**: Implementing safety measures helps chemical plants comply with regulations and standards, avoiding potential legal penalties and reputational damage. **Operational Efficiency**: Safety measures can improve the efficiency of operations by minimizing downtime and costs associated with accidents and repairs. **Community Safety**: By ensuring a safe working environment, safety measures contribute to the overall safety of the community, reducing the risk of accidents that could affect the local population.

(Explain any three factors)

13.How does alloying help in reducing corrosion
Alloying elements in metals and alloys play a crucial role in reducing corrosion by forming protective layers and enhancing the metal's ability to resist degradation.
Chromium: Forms a protective oxide layer on metals, significantly enhancing their resistance to corrosion

14.What is Bayers Process, and why is it important in the production of alumina
The **Bayer process** is the principal industrial means of refining bauxite to produce alumina (aluminium oxide). Bayers process is important because it is used to produce alumina, which is essential in the production of aluminium metal.

15.Write any two applications of fullerenes
Biomedical Applications- Drug Delivery- due to its ability to interact well with cells and also pass through cell membranes. **Catalysis-** Fullerene-Transition metal complex can be used as a catalyst to break down H_2O into H_2 and O_2, **Lubricants**-Due to its spherical shape, fullerene has been exploited as a solid lubricant, The small spheres can provide lubrication by rolling over one another and rolling between the two solids. **Electronics-**Fullerenes are often used in preparation of electronic devices, semiconductors etc. Preparation of **batteries** as charge carriers. ***(Any two applications)***

16. Compare the size of any three nanomaterials and classify them
Nanomaterials with a minimum of **one dimension** in the nanoscale are known as nanolayers. Examples are **thin films and surface coatings**. Nanomaterials with **two dimensions** in the nanoscale are referred to as **nanotubes or nanowires.** Examples of these include **carbon nanotubes and carbon.** Nanoparticles which have the size ranges from 1 to 100 nm in **three dimensions .**These are tiny aggregates of atoms but smaller than their crystals but bigger than molecules. Example- **colloids and precipitates.(Gold , TiO_2, Silver)**

PART C

Short essay type questions, answer any 4, each question carries 7 marks

17. Discuss the significance of purity in fine chemicals and the industries where the fine chemicals are typically used.
Beyond the scope of the syllabus. Give credit to all

18.Evaluate the impact of process parameters (temperature,pressure, catalyst)on the rate and selectivity of a chemical reaction

Process Parameter	**Effect on Reaction Rate**	**Effect on Selectivity**
Temperature	• Increases rate by increasing molecular kinetic energy.• Higher temperature → more molecules cross activation energy barrier (Arrhenius).	• Low temperature → favors kinetic products.• High temperature → favors thermodynamic products.• Very high temperature can increase side reactions and reduce selectivity.
Pressure	• Higher pressure increases reaction rate for gas-phase reactions by increasing gas concentration and collision frequency.	• Favors product side with fewer gas molecules (Le Chatelier's principle).• Can promote selective adsorption on catalytic surfaces, affecting product distribution.
Catalyst	• Lowers activation energy and accelerates reaction rate without altering equilibrium.• Provides an alternative pathway.	• Determines which pathway is favored (reaction-specific active sites).• Enables high product selectivity (e.g., shape-selective catalysis, selective hydrogenation)

Give credit to reasonable attempt

19.Describe the electrochemical process of rusting in detail. Explain the reactions that occur at the anode and cathode during rust formation and the conditions necessary for this process.

Rusting is a chemical process where iron reacts with oxygen and water to form iron oxide (rust). The electrochemical theory of rusting describes the process in which iron reacts with water and oxygen to form rust, which is a form of iron(III) oxide. The reaction is electrochemical in nature as it involves a transfer of electrons.When iron comes into contact with water or moist air or an aqueous environment, it undergoes a redox reaction. Two distinct electrochemical reactions occur in the overall rusting process.

1. Anodic Reaction: At the anode, iron gets oxidized by losing electrons and is transformed into iron(II) ions. $Fe(s)$--⊠$Fe^{2+} + 2e^-$

2. Cathodic Reaction: Simultaneously, at the cathode, oxygen from the air or water gains electrons to be reduced by reacting with water (moisture) to form hydroxide (OH^-) ions.

$O_2(g) + 2H_2O(l) + 4e^- \longrightarrow 4OH^-$

These iron(II) ions then further react with the formed hydroxide ions to form a greenish precipitate which is iron(II) hydroxide ($Fe(OH)_2$).

$Fe^{2+} + 2OH^- \longrightarrow Fe(OH)_2$

But the tendency of iron(II) hydroxide ($Fe(OH)_2$) is to undergo oxidation in the presence of dissolved oxygen from the air to form hydrated iron(III) oxide ($Fe_2O_3.xH_2O$) or rust.

For rusting to happen 1.**Presence of Water (Moisture):** Iron needs to be in contact with water, even if it's just humidity in the air. 2.**Presence of Oxygen:** Oxygen from the air reacts with iron in the presence of water.

20. Discuss the significance of regenerated cellulose (viscose) in textile industry.

1. Versatility: Viscose can be made into fabrics that feel like cotton, silk, or wool. This makes it useful for a wide range of clothing—like sarees, shirts, dresses, and even home textiles such as curtains and bed sheets.**2. Comfort:** Viscose is soft, breathable, and absorbs moisture well. In India's hot and humid climate, these properties make viscose fabrics comfortable to wear.3**. Cost-effective:** It's cheaper to produce than natural silk or cotton, so it's popular for making affordable, stylish clothes.**4. Dyeing:** Viscose takes dyes very well, so fabrics can be made in bright, vibrant colours that don't fade easily.**5. Blending:** It blends easily with other fibres like cotton or polyester, improving the texture, strength, and appearance of fabrics.**6. Biodegradability:** Since viscose is made from natural cellulose (usually wood pulp), it's more biodegradable than many synthetic fibres like polyester.
(any four points)

21. Explain the underlying principles of Chemical vapour deposition (CVD) and describe its process in detail
Chemical vapor deposition is a bottom up approach used process for producing high-quality thin films and nanoparticles by chemical reactions of gaseous precursors on heated substrates.
commonly used in the semiconductor industry to create thin films of materials such as silicon, carbo nand various metal oxides. CVD is favored for its ability to produce highpurity and uniform coatings, making it essential in applications ranging from microelectronics to nanotechnology.
Process of CVD
Preparation-The substrate (like a silicon wafer) is cleaned and placed inside a reaction chamber.**Introduction of Reactant Gases** Gaseous chemicals (precursors) are introduced into the chamber. Common examples: silane (), ammonia (), hydrogen (), etc. **Chemical Reaction** The chamber is heated (sometimes using plasma or lasers).The reactant gases decompose or react at the hot substrate surface, forming a solid material (the film) and gaseous by-products.**Film Formation** The solid product deposits as a thin, uniform layer on the substrate.The by-product gases are continuously removed from the chamber.

22. What are the limitations of top down approach in nanomaterial synthesis. Provide examples of industries where this method is commonly used.

Beyond the scope of syllabus, Give credit to reasonable attempt. Give credit to explanation of top down approach.

Top down approach involves breaking down of larger bulkier materials into smaller nanoparticles through various methods
Limitations 1.Cost: The initial setup and operation of a top-down approach require significant capital investments, making it less suitable for mass production.2. Size Control: This method often lacks fine control over particle size distribution and morphology, which can lead to defects in the final product. 3.Contamination: The milling process can introduce contaminants into the material, affecting its purity and quality. 4.Processing Time: Prolonged milling times may induce structural defects that alter material properties unpredictably. **Industries–** Electronics, Medicines, Energy storage etc.

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